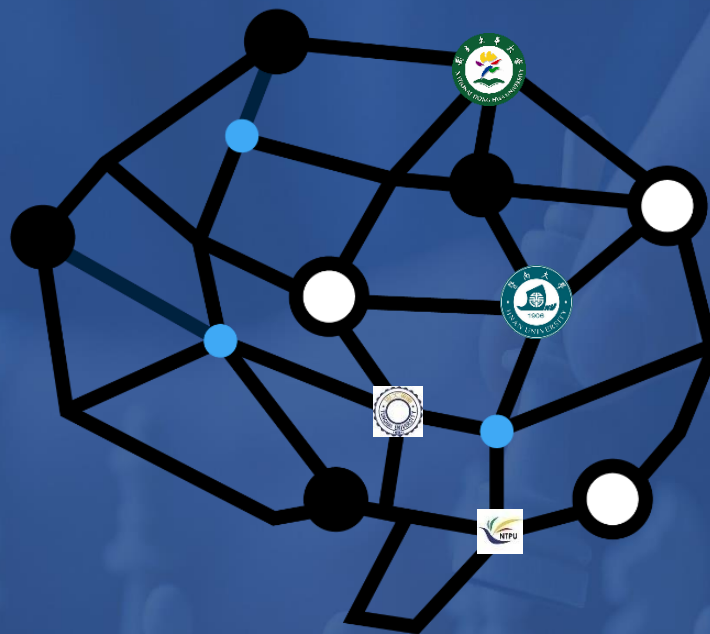




Chap. 1 Introduction

課程：人工智慧導論

授課教師：周信宏

Outline

- What is AI?
- AI history



I

01

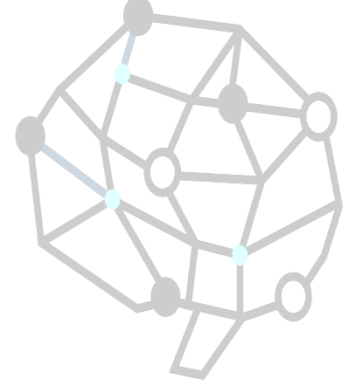


What is AI?

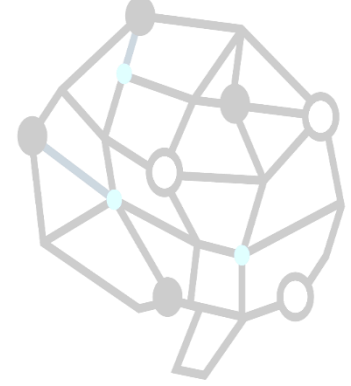
Information of chapter

What is AI?

- Systems that...
 - Thinking humanly?
 - Thinking rationally?
 - Acting humanly?
 - Acting rationally?



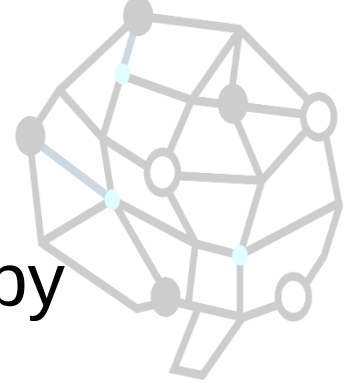
Thinking Rationally: Laws of Thought



- Aristotle:
 - correct arguments/thought processes
- Logic:
 - notation and rules of derivation for thoughts
- Direct line thought **mathematics** and **philosophy** to **modern AI**

Problems

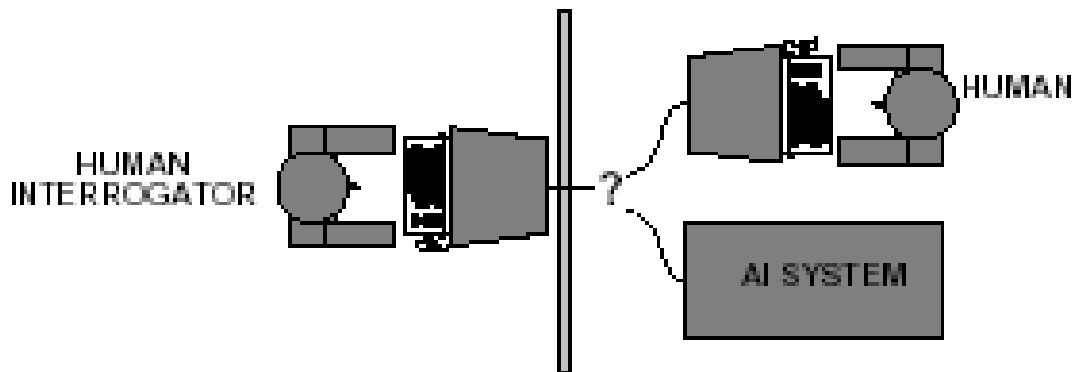
- Not all intelligent behavior is mediated by logical deliberation
- What is the purpose of thinking?



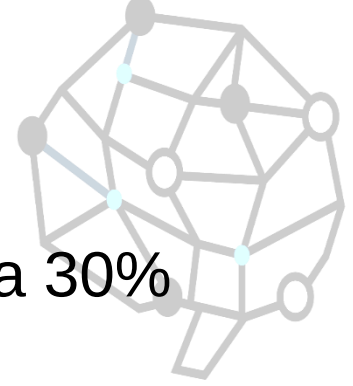
Acting Humanly: The Turing Test(1/2)



- Turing(1950) “Computing machinery and intelligence”
- “Can machines think?”; “Can machines behave intelligently?”
- Operation test for intelligent behavior:

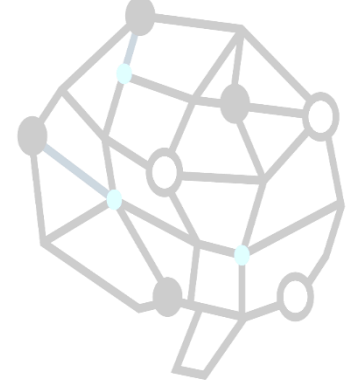


Acting Humanly: The Turing Test(2/2)



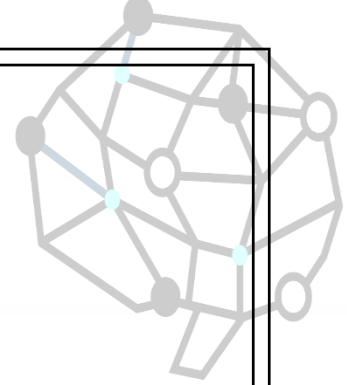
- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Anticipated all major arguments against AI in following 50 years
- Suggested major components of AI: knowledge, reasoning, language understanding, learning
 - Watson

Acting Rationally



- Rational behavior: doing the right thing
- The right thing
 - That which is expected to maximize goal achievement, given the available information
- Aristotle: Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good

Rational Agents



An agent is an entity that perceives and acts

Abstractly, an agent is a function from percept histories to actions: $f: \mathcal{P}^* \rightarrow \mathcal{A}$

For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

Caveat: computational limitations make perfect rationality unachievable → design best program for given machine resources

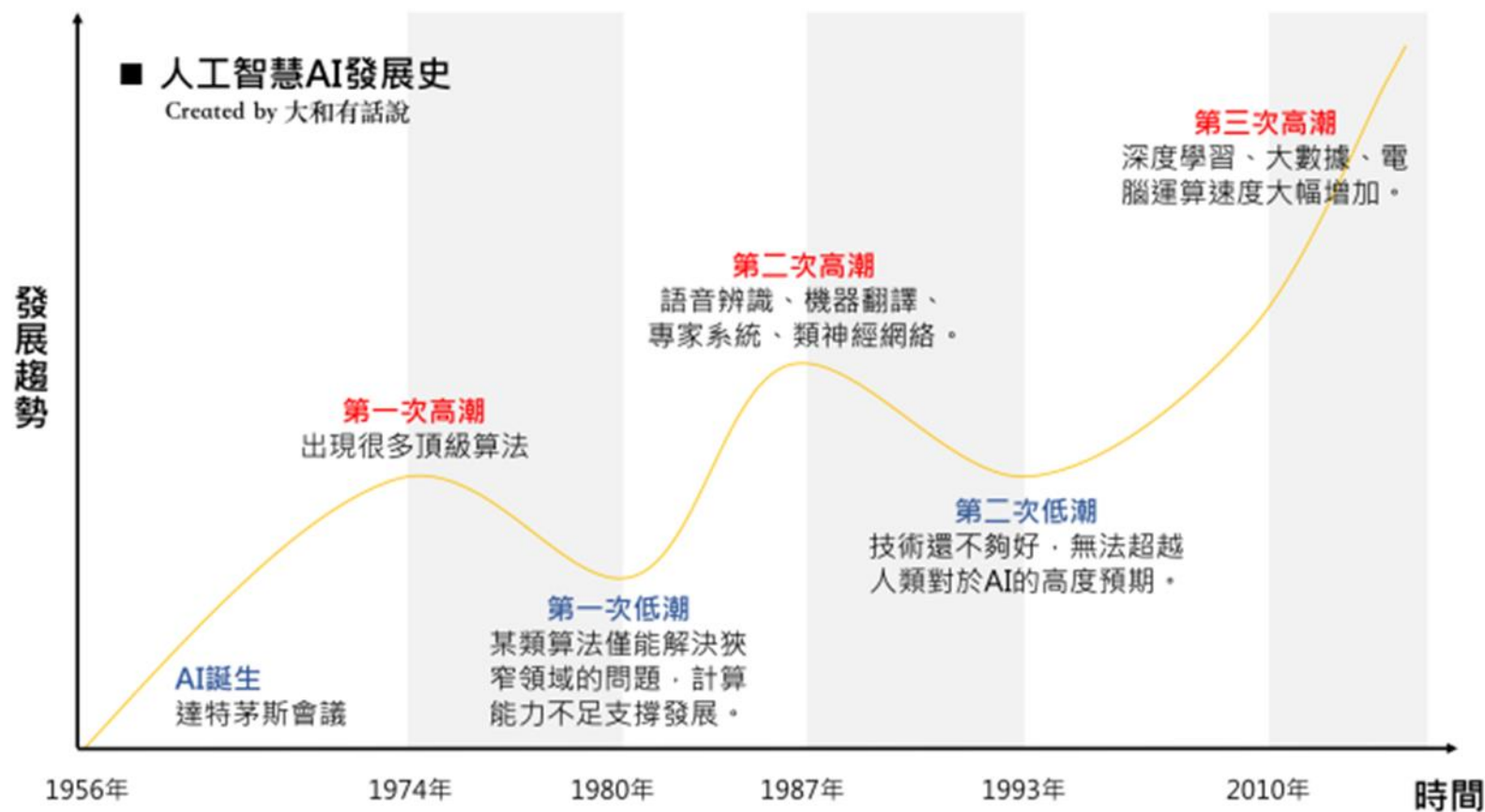
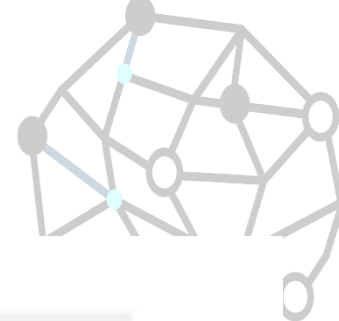
I

02

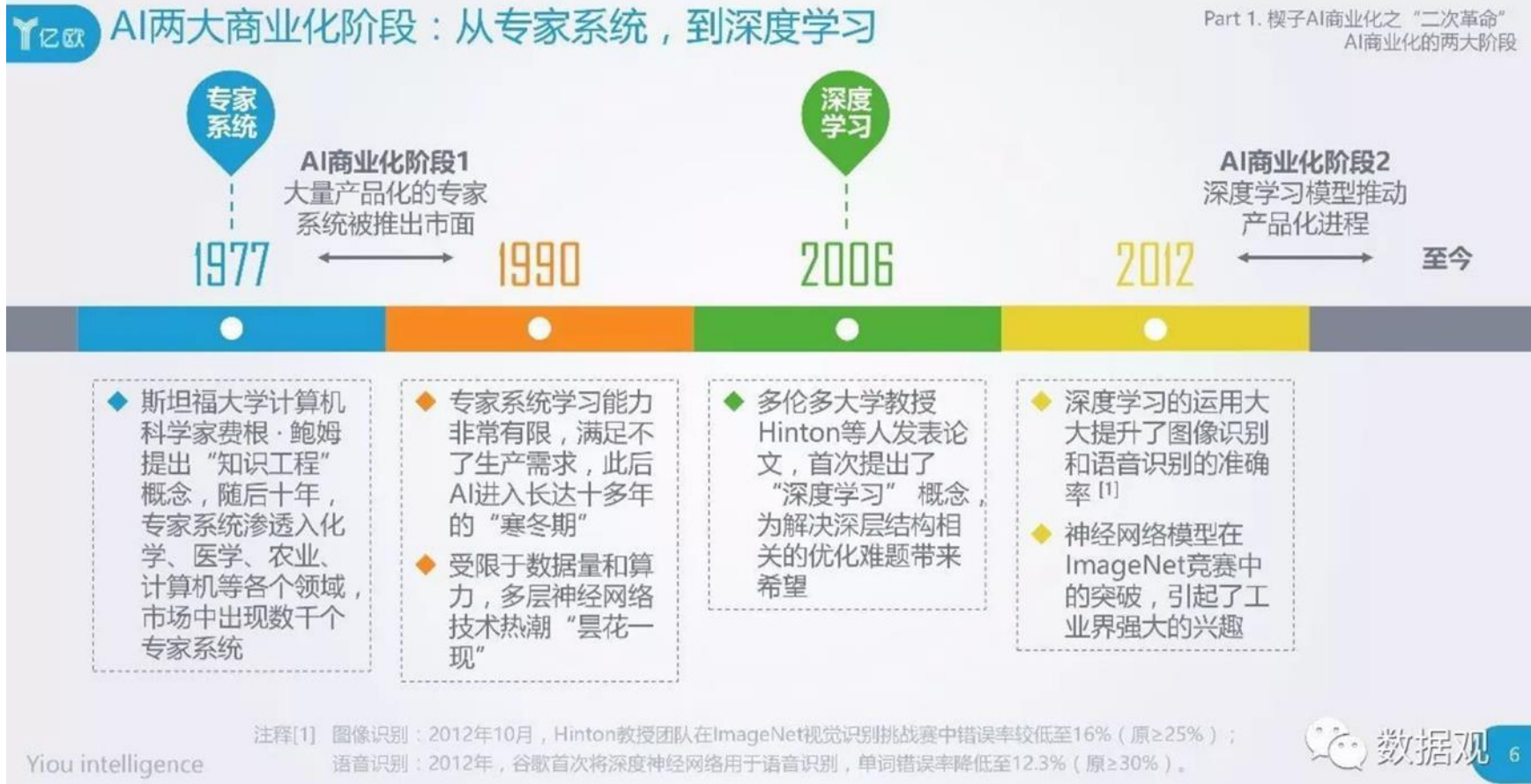


AI History

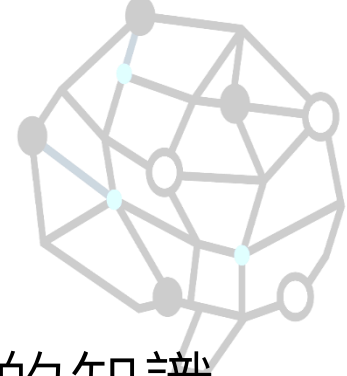
AI發展史



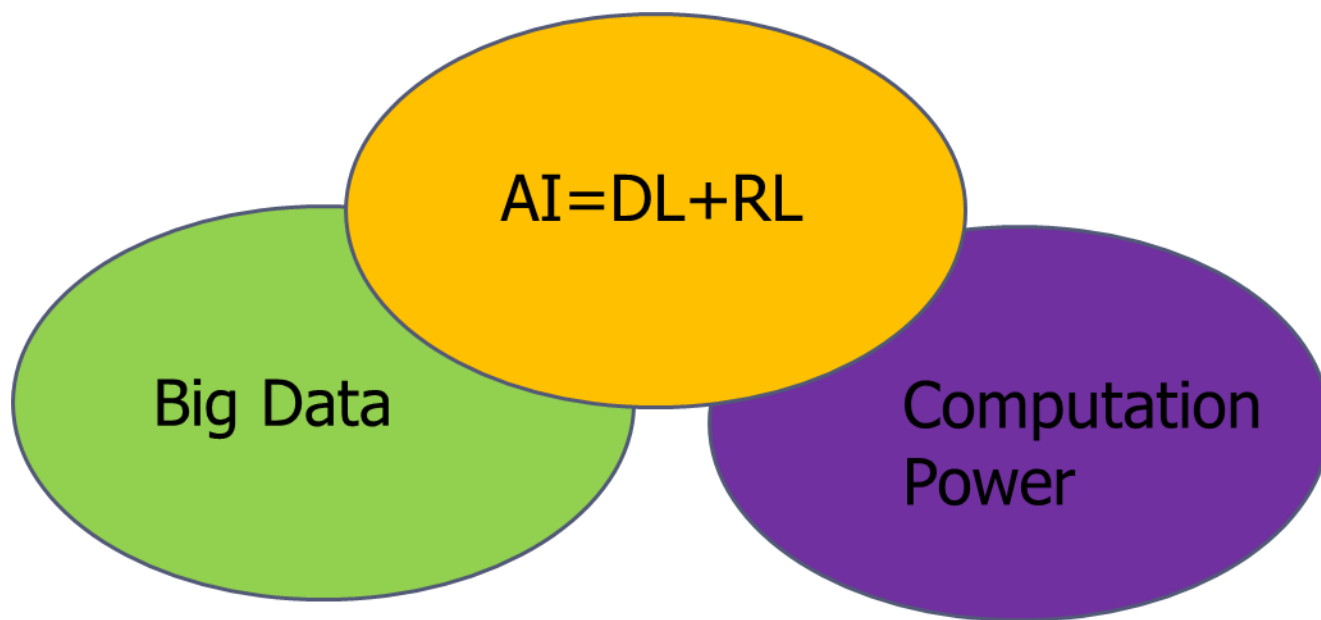
2017中国AI投资市场研究报告



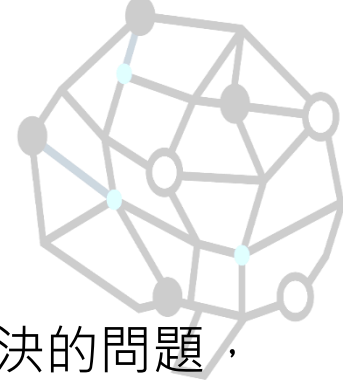
AI=DL+RL(by David Silver)



- Deep Learning : 儲存知識
- Reinforcement Learning : 搜尋得到更好的知識



人工智慧三層次定義



- 弱人工智慧(狹隘人工智慧)
 - 基本上就是希望電腦能解決某個需要高度智力才能解決的問題，而不要它跟人類一樣有全面智慧解決各式各樣不同的問題。
例如 **Google AlphaGo**、**IBM Watson** 或自動駕駛，就是弱人工智慧的例子。
- 強人工智慧(泛人工智慧)
 - 強人工智慧要求電腦的智慧需要更全面廣泛，需要有推理、學習、規劃、語言溝通、知覺等能力，擁有這些能力的電腦才有可能展現出全面性的智慧、跟人類並駕齊驅。
- **John Searle (強人工智慧假說) (strong AI hypothesis)**
 - 在這個層次，電腦需要擁有跟人類一樣的「心靈」，需要認知自我並可以跟人類一樣思考。

人工智慧 (AI)

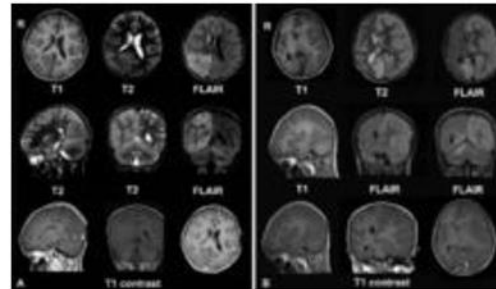
-人類社會的下一波巨大變革<「我國的ai科研戰略」簡報>



Source: [flickr](#)



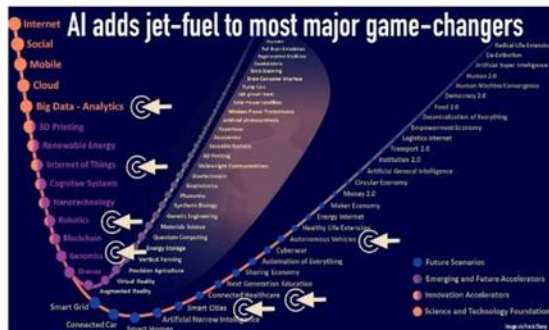
introducing
amazon echo
Always ready, connected,
and fast. Just ask.



Source: [wikipedia](#)



Source: [wikipedia](#)



Source: [flickr](#)

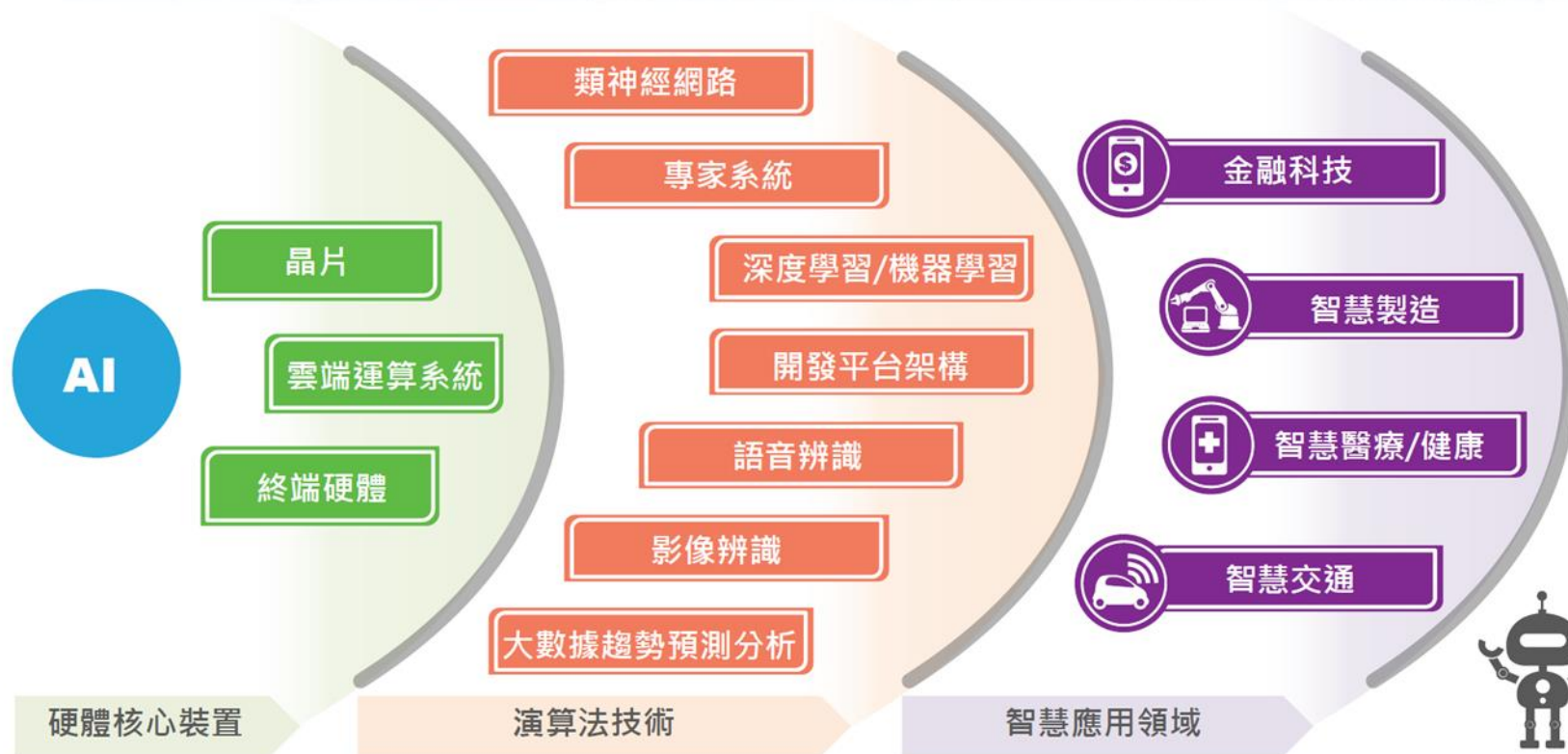


Source: [freegreatpicture.com](#)



Source: [wikipedia](#)

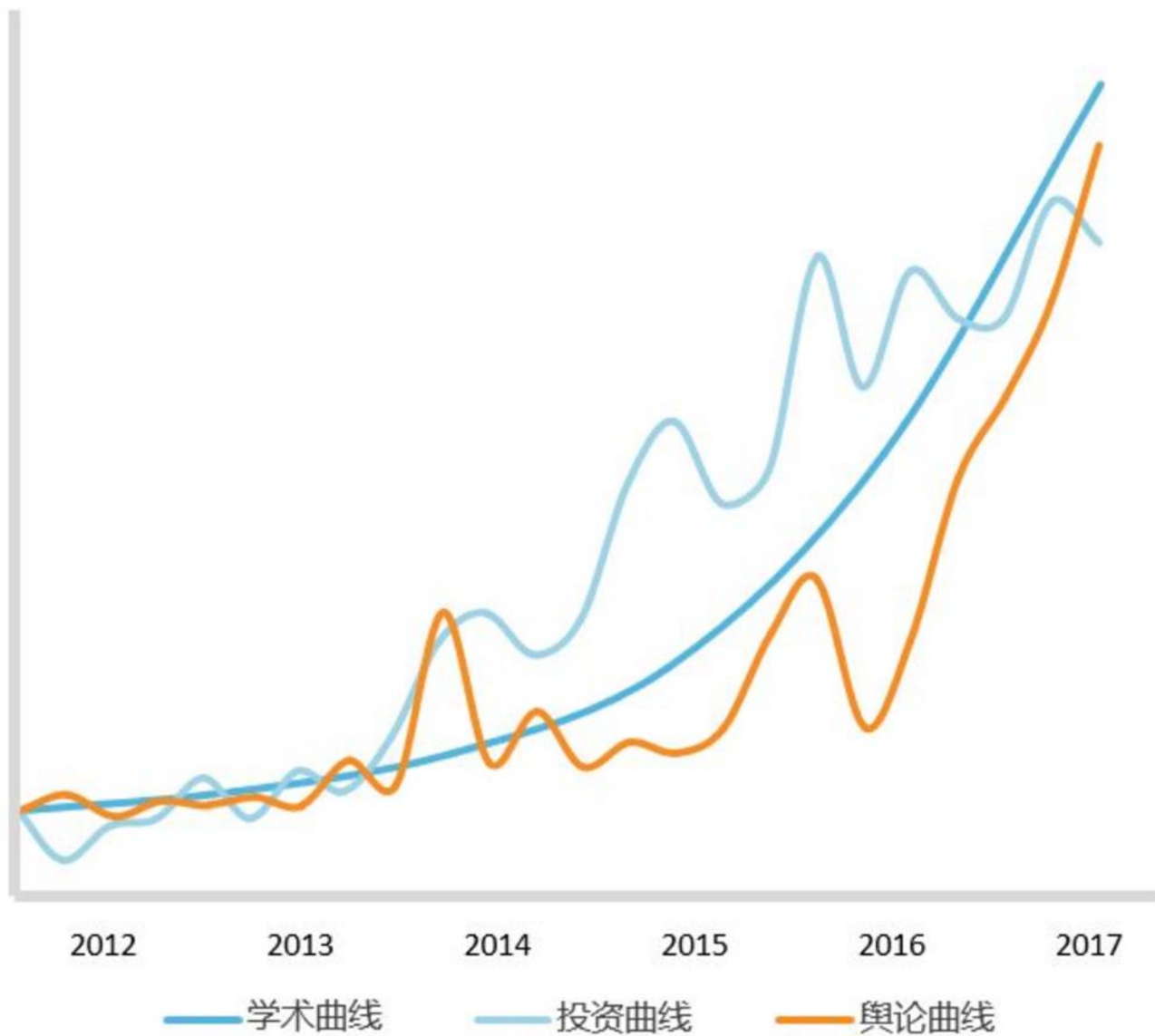
人工智慧涵蓋面向<「我國的AI科研戰略」簡報>



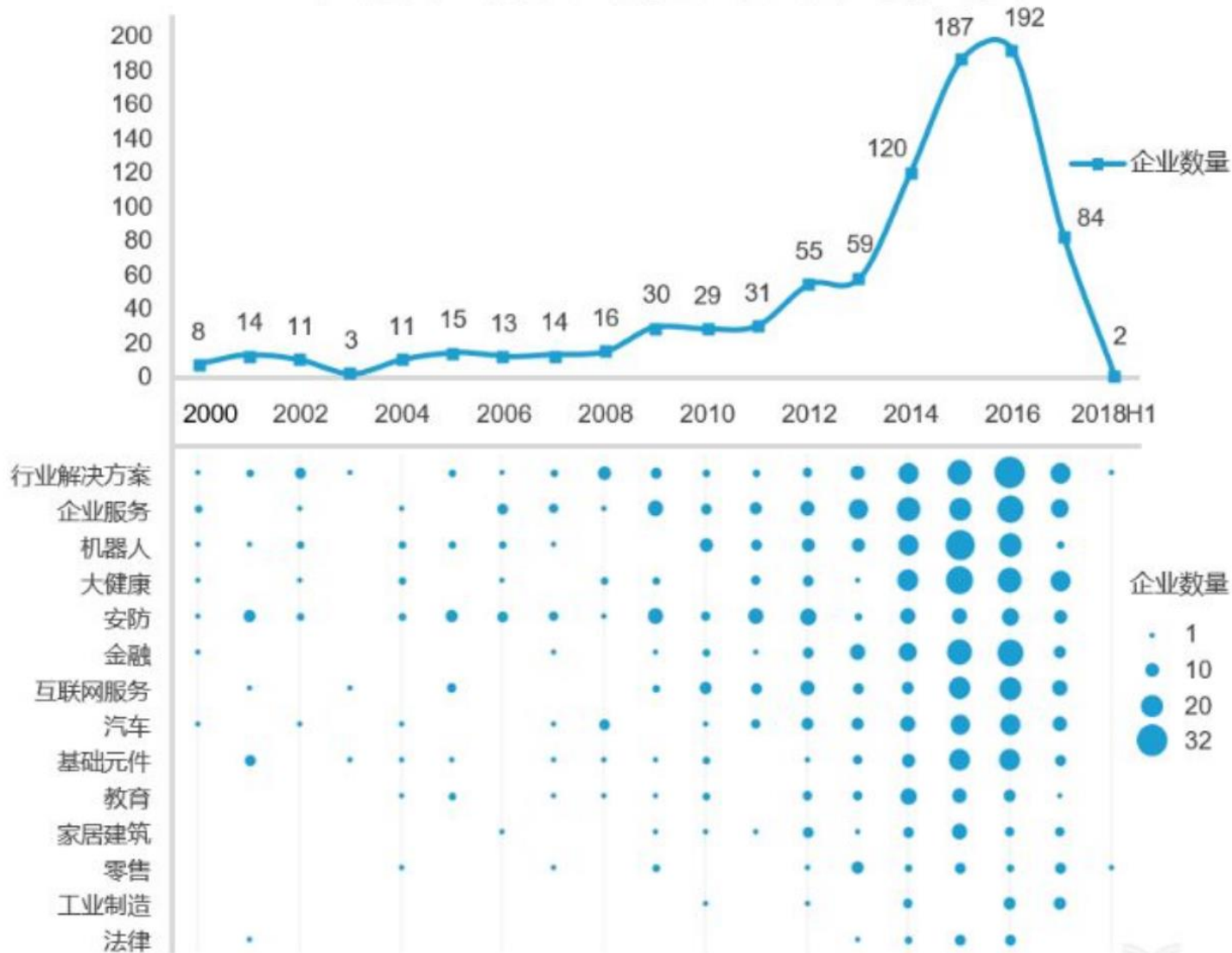
亿欧智库：中国人工智能学术、投资与舆论曲线



增长趋势



亿欧智库：中国人工智能企业诞生历程与行业分布



来源：亿欧智库人工智能企业数据库

亿欧 (www.iyiou.com)

A blue-tinted background image showing a hand moving a chess piece on a chessboard. The text "THE END." is centered over the image.

THE END.
